

Radiation distribution

The Rayleigh-Jeans law failed to describe the ranges of wavelength emitted at all but the longest wavelengths, with energy increasing infinitely at smaller and smaller wavelengths.

Experiments have shown that black bodies emit radiation across a spread of wavelengths that becomes narrower in range, with a shorter peak wavelength, as their temperature increases. Cool objects emit primarily long infrared waves, then glow red, white, and eventually blue and ultraviolet as they get hotter. An equation called the Rayleigh-Jeans law predicted the pattern of radiation for cooler objects, but also suggested that emission would increase exponentially toward higher temperatures and eventually become infinite, which was called the “ultraviolet catastrophe.”

